

자바 실습 문제 - 참조타입 (정답)

1. 정답

설명: '=='는 참조(주소값)를 비교하고, equals()는 실제 문자열 값을 비교합니다.

2. 정답

```
Scanner scanner = new Scanner(System.in);
String inputString = scanner.nextLine();
String inputStringLow = inputString.toLowerCase().replace(" ", "");

int[] arrays = new int[26];
for (int i=0;i<inputStringLow.length();i++) {
    int ch = (int) inputStringLow.charAt(i) - (int) 'a';
    arrays[ch] += 1;
}
int maxNum = 0;
char maxChar = 0;
for (int i=0;i<arrays.length; i++) {
    if (maxNum < arrays[i]) {
        maxNum = arrays[i];
        maxChar = (char) (i + (int) 'a');
    }
}
System.out.println("가장 많이 등장한 문자: " + maxChar + " (" + maxNum + "회)");
```

3. 정답

```
int[][] multiArray = new int[3][3];
System.out.println("배열");
for (int i=0; i < multiArray.length; i++) {
    for (int j=0; j<multiArray[i].length; j++) {
        multiArray[i][j] = (int) (Math.random() * 100);
        System.out.printf("%02d: (%02d, %02d) ", multiArray[i][j], i, j);
    }
    System.out.println();
}
maxNum = 0;
int row = 0;
int col = 0;
for (int i=0; i < multiArray.length; i++) {
    for (int j=0; j<multiArray[i].length; j++) {
        if (maxNum < multiArray[i][j]) {
            maxNum = multiArray[i][j];
        }
    }
}
```

```

        row = i;
        col = j;
    }
}
System.out.printf("최대값\n%02d: (%02d, %02d)\n", maxNum, row, col);

```

4. 정답

```

int[] colSum = {0, 0, 0};
double[] colAvg = {0, 0, 0};
for (int i=0; i < multiArray.length; i++) {
    for (int j=0; j<multiArray[i].length; j++) {
        colSum[i] += multiArray[j][i];
        colAvg[i] += (double) multiArray[j][i] / multiArray[0].length;
    }
}
System.out.printf("총합: %d, %d, %d\n", colSum[0], colSum[1], colSum[2]);
System.out.printf("평균: %2.2f, %2.2f, %2.2f\n", colAvg[0], colAvg[1], colAvg[2]);

```

5. 정답

```

enum SeatStatus {EMPTY, RESERVED, BLOCKED}
SeatStatus[] seatStatuses = new SeatStatus[10];
System.out.println("초기상태: ");
for (int i=0; i<seatStatuses.length; i++) {
    int rand = (int) (Math.random()*3);
    switch (rand) {
        case 0 -> seatStatuses[i] = SeatStatus.EMPTY;
        case 1 -> seatStatuses[i] = SeatStatus.RESERVED;
        case 2 -> seatStatuses[i] = SeatStatus.BLOCKED;
    }
    System.out.printf("[%s]", seatStatuses[i].toString().charAt(0));
}
System.out.println();
boolean success = false;
do {
    int inputInt = scanner.nextInt();
    System.out.print(inputInt + "번 좌석 예약 ->");
    if (seatStatuses[inputInt].equals(SeatStatus.EMPTY)) {
        seatStatuses[inputInt] = SeatStatus.RESERVED;
        System.out.print("예약 성공\n");
        success = true;
    } else if (seatStatuses[inputInt].equals(SeatStatus.RESERVED)) {
        System.out.print("예약 실패 (이미 예약됨)\n");
    } else if (seatStatuses[inputInt].equals(SeatStatus.BLOCKED)) {
        System.out.print("예약 실패 (차단됨)\n");
    }
} while (!success);

```

```
    }  
} while(!success);  
  
System.out.println("최종 상태: ");  
for (SeatStatus seatStatus : seatStatuses) {  
    System.out.printf("[%s]", seatStatus.toString().charAt(0));  
}  
System.out.println();
```